

CSSCM 2026 COHORT

Numerical Python Practice Exercise

Multicomponent Spectrophotometric Analysis Using NumPy

Director	Elliot S. Menkah (PhD)
Tutors	Emmanuel Oppong (MPhil); Obed Adu-Gyamfi (BSc)

Practice Question

A chemist is analysing a solution containing three absorbing species, **X**, **Y** and **Z**. The mixture is measured at three different wavelengths. After applying the Beer-Lambert relationship and the known response coefficients of the three species, the following simultaneous equations are obtained:

$$3C_X + C_Y + 2C_Z = 7.4$$

$$C_X + 4C_Y + C_Z = 5.9$$

$$2C_X + 2C_Y + 3C_Z = 8.5$$

where C_X , C_Y and C_Z are the unknown concentrations of the three chemical species.

Matrix Representation

The system may be represented in the general form:

$$\mathbf{EC} = \mathbf{A}$$

$$\mathbf{E} = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 4 & 1 \\ 2 & 2 & 3 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} C_X \\ C_Y \\ C_Z \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 7.4 \\ 5.9 \\ 8.5 \end{bmatrix}$$

Your Task

1. Import NumPy using the standard alias `np`.
2. Create the 3 x 3 coefficient matrix **E** as a NumPy array.
3. Create the experimental vector **A** as a NumPy array.
4. Display the shape and number of dimensions of **E**.
5. Solve the system **EC = A** using `np.linalg.solve(E, A)`.
6. Store the result in a NumPy array called **C**.
7. Print C_X , C_Y and C_Z separately to four decimal places.
8. Verify the solution by calculating $\mathbf{E} @ \mathbf{C}$.
9. Compare the calculated vector with the original experimental vector **A**.
10. Use `np.allclose()` to make Python report whether the two vectors agree numerically.

Useful NumPy Syntax

```
import numpy as np

# Create an array
matrix = np.array([...])

# Solve a linear system
solution = np.linalg.solve(matrix, vector)

# Matrix-vector multiplication
```

```

check = matrix @ solution

# Compare numerical arrays
np.allclose(check, vector)

```

Important: The syntax above is a reference only. You must construct the correct arrays from the chemical system yourself.

Expected Output Structure

```

=====
MULTICOMPONENT CHEMICAL ANALYSIS
=====

Coefficient matrix E:
[...]

Experimental vector A:
[...]

Concentration vector C:
[...]

Species X concentration = _____
Species Y concentration = _____
Species Z concentration = _____

Calculated experimental vector:
[...]

Original experimental vector:
[...]

Solution verified: _____
=====

```

Questions for Interpretation

- What does each *row* of the coefficient matrix represent?
- What does each *column* of the coefficient matrix represent?
- Why should $E @ C$ reproduce the experimental vector if the solution is correct?
- Why is `np.allclose()` generally more appropriate than directly using `==` when comparing calculated floating-point arrays?

Submission

Save your program as `multicomponent_analysis.py` and run it in the Ubuntu terminal using:

```
python3 multicomponent_analysis.py
```

Submit the Python script together with a screenshot of the successful terminal output. Do not submit manually calculated values in place of the NumPy calculation.